Dear Editor,

Please consider our manuscript "Silent but deadly? Physics says seldom, memory says mostly" for publication as a Research Article in *Journal of the Royal Society Interface*.

The belief that a silent release of flatus smells worse than an audible one is recorded in many languages, yet no mechanistic test of it exists. The closest prior work is a conference abstract on the physics of flatulence (Ancalle et al. 2021). We follow one passage through a single chain: chemistry, sound, dispersion, perception and memory. Three results reach beyond the topic.

1. **Routing by the human convective boundary layer.** In simulations of a sofa, a standing pair and a closed lift, the warm air rising along a body takes a slow release next to it and lets a fast one pass, so the momentum of the release, relative to that of the boundary layer, decides which nearby nose it reaches. The same set-up reproduces measured breathing-zone intake from a source at a seated manikin's groin within a factor of 1.5. This bears on any near-body source, such as respiratory, skin and clothing emissions in shared indoor spaces.

2. **A peak bound for linear transport.** We prove that a slower release of the same substance cannot give a higher peak concentration, in any room or flow. Only the non-linear near field lets it win, by a factor bounded by a velocity ratio, which we call the Silencer number.

3. **Selection across two senses.** With passages as emitted and settings weighted by how often people share them, a bystander finds the silent passage worse in 0.5% of pairs and the audible one in 6.9%; only in a closed lift, for the person behind the releaser, are the two about equal. Yet because a silent passage is noticed only if it is smelled, a remembered silent passage seems the worse one in 88% of comparisons. When an event can be noticed through either of two senses, selection on noticing alone can create a false sensory belief. The explanation makes testable predictions: releasers should hold the belief less firmly than bystanders, and people without a sense of smell should never remember a silent passage.

The work joins fluid dynamics, chemistry, acoustics, sensory science and cognition. We think it suits *Interface* because each part can be read on its own, and because the question shows how a physical and a cognitive explanation can be separated. The headline is checked against the scene weights, the observer models, large-eddy peak factors, mesh refinement and clothing; its direction holds in every case but one adverse pairing of peak factors, and the remembered share stays above 80% in all of them.

Because the work spans several fields, our suggested reviewers cover them one each: indoor plume and ventilation modelling (H. C. Burridge), the human thermal plume and CFD validation (I. Năstase), turbulent puffs and jets (S. S. Diwan), olfactory perception (A. Oleszkiewicz) and judgement from selective samples (H. Alves). We have no collaboration with any of them.

Equations and proofs are in appendices, and technical details are in the electronic supplementary material. Code and processed data are openly available at `https://github.com/ccy5123/sbd`; an archived release with a DOI (Zenodo) will accompany the revised manuscript. [**A preprint is posted on engrXiv (DOI . . .).**] The results have not been reported in the media. The manuscript is not under consideration elsewhere. AI-assisted technologies (Claude Code, Anthropic) were used for code, simulations, literature search and drafting, as declared in the manuscript.

Yours sincerely,

Chan Young Joe
Independent Researcher, Incheon, Republic of Korea
ORCID 0009-0007-5822-6714